California Sun: Demographics of Large-Scale Solar Farms in California

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RESEARCH QUESTION

• What demographic characteristics are associated with the location of large-scale solar facilities in California?

LITERATURE REVIEW

• Brewer et al. (2015) notes that suitable sites for large-scale solar projects depend not only on physical constraints, but also heavily on community preferences and social factors.

• O'Shaughnessy et al. (2022) points out utility-scale solar and wind project locations in the US are chosen primarily by factors such as open land and grid access, which tend to overlap with sparsely populated and typically lower income areas.

• Hernandez et al. (2015) emphasizes that large-scale solar development often overlaps with ecologically sensitive desert landscapes, creating conflicts in land-use and negatively impacting the surrounding environment.

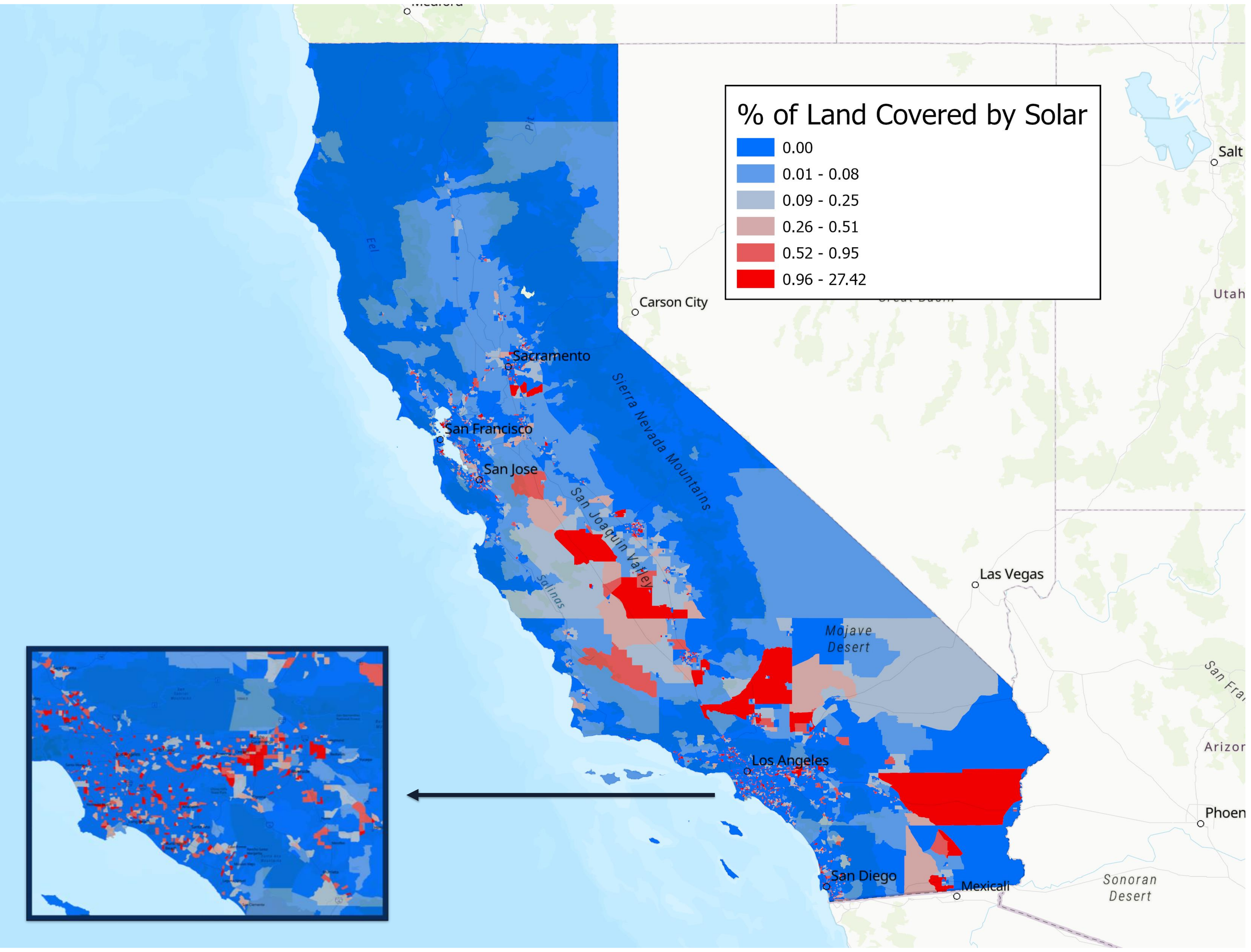
METHODS & DATA SOURCES

- ArcGIS Pro 3.6
- OLS, spatial lag, and spatial error regression

- Census tract level demographic data from the US Census Bureau's 2019-2023 American Community Survey (ACS) five-year estimates
- Solar footprints in California from the California Energy Commission in 2025, which includes rooftop solar, parking lot solar, and ground solar all of which are roughly larger than 1 acre

Large-scale solar facility (American Public Power Association 2017)

PERCENTAGE LAND COVERED BY LARGE-SCALE SOLAR BY CENSUS TRACT

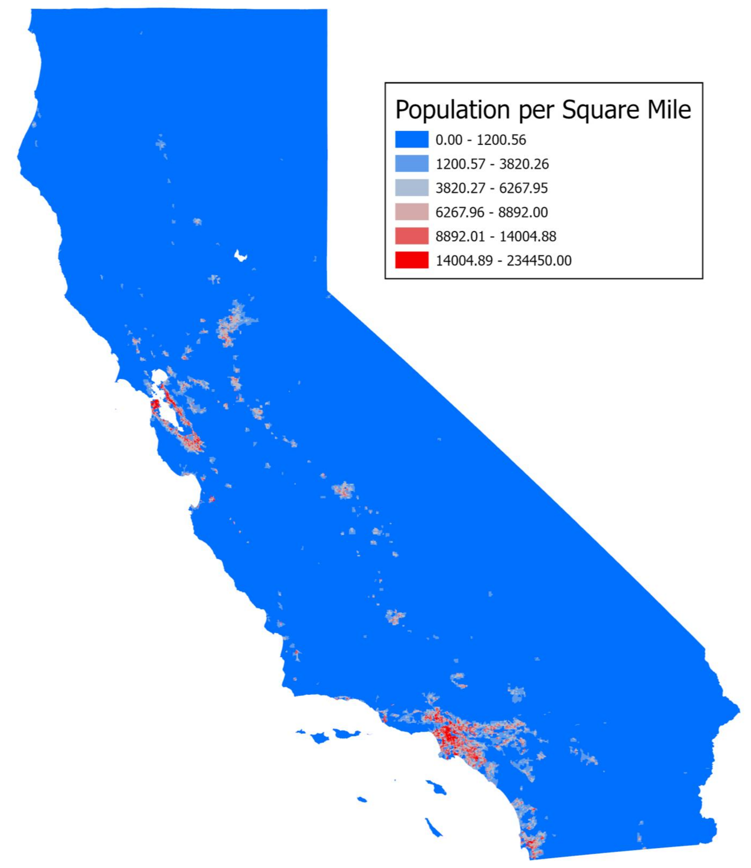


ANALYSIS

	coef	std err	t	P>|t|
Intercept	-0.010	0.010	-0.998	0.3181
Median_Household_Income	-0.019	0.013	-1.520	0.1285
Pop_per_Square_Mile	-0.313	0.012	-25.568	0.0000
Homeownership_Percent	-0.129	0.014	-8.990	0.0000
Percent_Unaffordable_Rent	-0.026	0.011	-2.481	0.0131

AIC: 24362.9578
Lambda: 0.4464
OLS Adj R-squared: 0.063

Spatial error diagnostics and coefficients for solar coverage by census tract in California from 2019-2023. The variables are median household income in US dollars from 2019-2023 (USCB n.d.), population per square mile from 2019-2023 (USCB n.d.), and percentage of rent that is deemed unaffordable by an index from 2019-2023 (USCB n.d.).



Population per square mile by California census tract from 2019-2023 (USCB n.d.)

CONCLUSION

- The low spatial error R-squared value of 0.063 indicates that most variation in solar development is driven by non-demographic factors, likely environmental, land-use, or infrastructure not captured in the American Community Survey data.
- Population density is the strongest demographic factor associated with the location of large-scale solar facilities in California, with census tracts containing more solar typically being far less populated.
- These results suggest that decisions around solar siting in California are driven primarily by physical and policy suitability rather than socioeconomic traits of local populations.

Solar development in rural area (PPIC 2022)

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